EE-451 Special project

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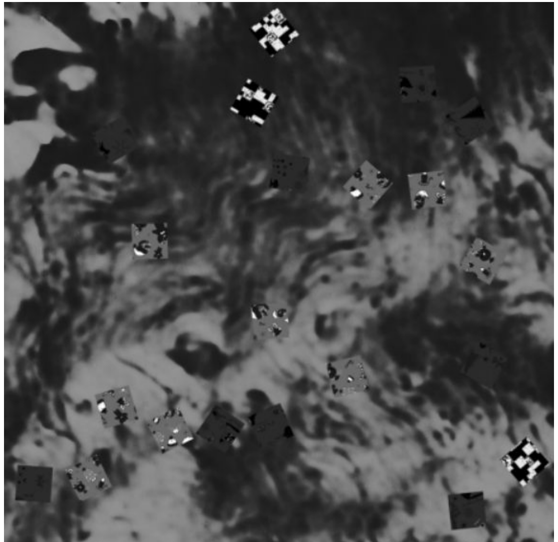
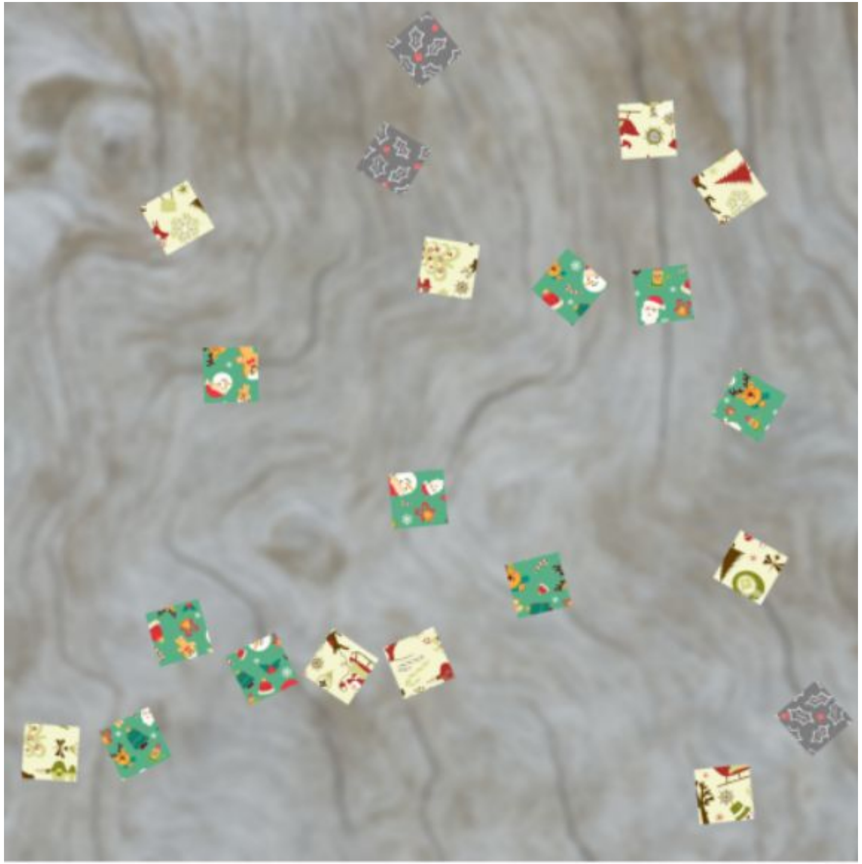
Segmentation: Channels

We used RGB + gray, saturation channels.

Bilateral filter: 75, 9.

Why not hue?



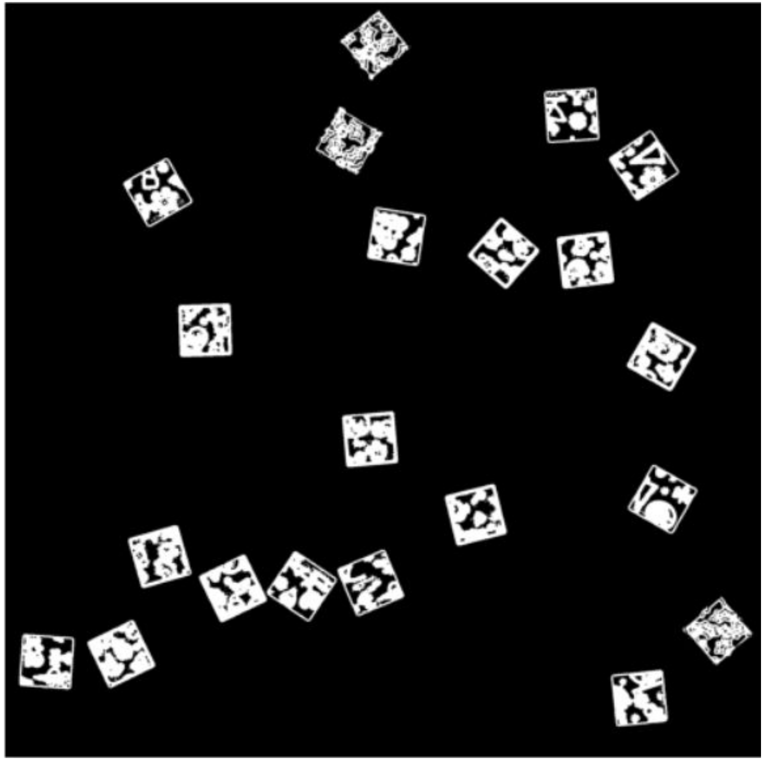


Segmentation: Thresholding and Morphology

Adaptive Thresholding + mixing masks

Closing: 1*1

Opening: 3*3



Segmentation: Finding Tiles

Find contours and detect based on area

Extract ROIs and expand each of them

Segmentation: Rotating Tiles

Fit polycons and extract vertices

Find slope and angle

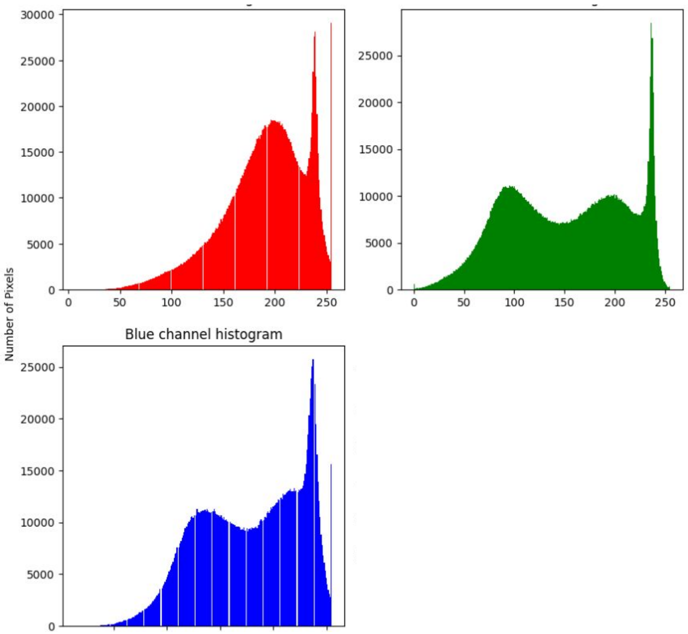
Find the middle point

Rotate

Feature extraction: color histogram

Compute histogram for each color channel

Concatenate obtained histograms



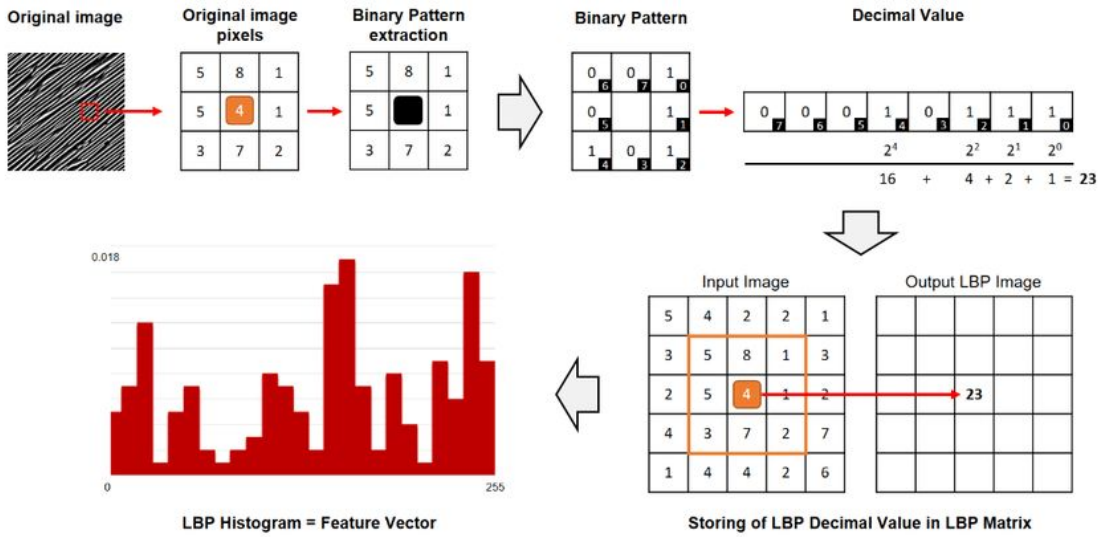
Feature extraction: Gabor features

Convolve puzzle piece with Gabor filters using all combinations of:

- Kernel size: 7
- Theta: [0, PI]
- Lambda: [5, 55, 105]
- Sigma: [10, 30, 50]
- Gamma: 0.5

Then compute Local Binary

Pattern histogram



https://www.researchgate.net/figure/Visualization-of-calculation-of-Local-Binary-Pattern-LBP-An-example-region-of-the_fig1_341341744

Feature extraction: putting it together

- Concatenate color histogram and Gabor features
- Normalization
- PCA (keep 99% of variance)

Clustering

DBSCAN:

- Does not require us to specify number of clusters in advance
- Can choose eps (radius around core points) parameter

Libraries

Pillow

Numpy

Skimage

OpenCV

Matplotlib